

Branch and bound for integer optimization

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Practice quiz

Consider the following integer optimization problem.

$$\min_{x \in \mathbb{N}^2} -13x_1 - 8x_2$$

subject to

$$x_1 + 2x_2 \leq 10,$$

$$5x_1 + 2x_2 \leq 20,$$

$$x_1, x_2 \in \mathbb{N}.$$

Solve the problem using the branch and bound algorithm.